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MIL-G-62496B(AT)
31 July 1992
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

GENERATOR, ENGINE ACCESSORY, 220 AMP

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, manufacture and acceptance requirements for the generator, engine accessory, 220 amp, shown on Drawing number 19207-12295268, herein referred to as the generator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils (Photosensitized Anodized Aluminum).
P-D-220	- Detergent, General Purpose.

MILITARY

MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-S-19500 JANTX	- Semiconductor Devices, Transistor.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461A Notice 4	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem, and Systems.
MIL-STD-462 Notice 3	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1275	- Electrical Circuit, 28 Volt DC, Transient Characteristics for Military Vehicles.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

19207-12295268	- Generator, Engine Accessory

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PUBLICATIONS

MRC-82304 MRC/SD-EP-82-027 - Nuclear Survivability Criteria for the IFV/CFV Systems (U), dated 9 September 1980, published by the U.S. Army Nuclear and Chemical Agency.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article (see 6.3).

3.1.1 Preproduction inspection (see 4.4.1 and 6.3.1). When specified (see 6.2), generators shall be subjected to preproduction testing. Production tooling is not required in the manufacture of preproduction samples.

3.1.2 Initial production inspection (see 4.4.2). When specified (see 6.2), generators shall be subjected to initial production inspection to demonstrate the adequacy and suitability of the supplier's processes and procedures in achieving the performance inherent in the design. Generators shall be produced with equipment and procedures used in production.

3.2 Materials. Materials shall be in accordance with the drawings, parts lists and other documents specified on Drawing 19207-12295268.

3.2.1 Dissimilar metals. The generator shall be designed and fabricated to minimize galvanic corrosion in dissimilar metal couples as specified in MIL-STD-889.

3.2.2 Corrosion resistance. Exposed functional parts shall be fabricated from corrosion-resistant materials or treated to prevent corrosion.

3.3 Design and construction. The design and construction of the generator shall be as specified herein and in accordance with Drawing 19207-12295268.

3.3.1 Dimensions. Dimensions and tolerances shall be in accordance with Drawing 19207-12295268.

3.3.2 Weight. The weight of the generator shall be not greater than 78 pounds (lbs).

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3.3.3 Protective coating. A protective finish shall be applied to the generator as specified on Drawing 19207-12295268.

3.3.4 Standard parts. Military standard parts shall be used wherever applicable.

3.3.5 Power transistor. The power transistor controlling the field current of the generator shall conform to MIL-S-19500.

3.3.6 Bearings. The generator shall have bearings that are open to lubrication, or, as an optional design, bearings that are sealed. If lubricative, the bearings shall be subject to the greasing conditions specified in 3.3.6.1.

3.3.6.1 Lubrication. The generator shall require no lubrication for the first 1000 hours of operation, or 2 years at a standstill, whichever occurs first. The lubricant shall be grease conforming to MIL-G-10924. Relubrication shall be possible.

3.3.7 Fan guard. The generator shall be provided with a guard to protect personnel from rotating fan disc.

3.3.8 Environmental stress screen (ESS) of the regulator. Unless otherwise specified by the procuring activity (see 6.2), ESS of the regulator shall be performed as part of the manufacturing process. Each assembly shall be stress screened in accordance with figure 1. The regulator shall be operated at nominal voltage for 15 cycles (see 6.4.1, 6.4.2 and 6.6).

3.4 Performance. Unless otherwise specified, the following conditions shall apply for inspection purposes:

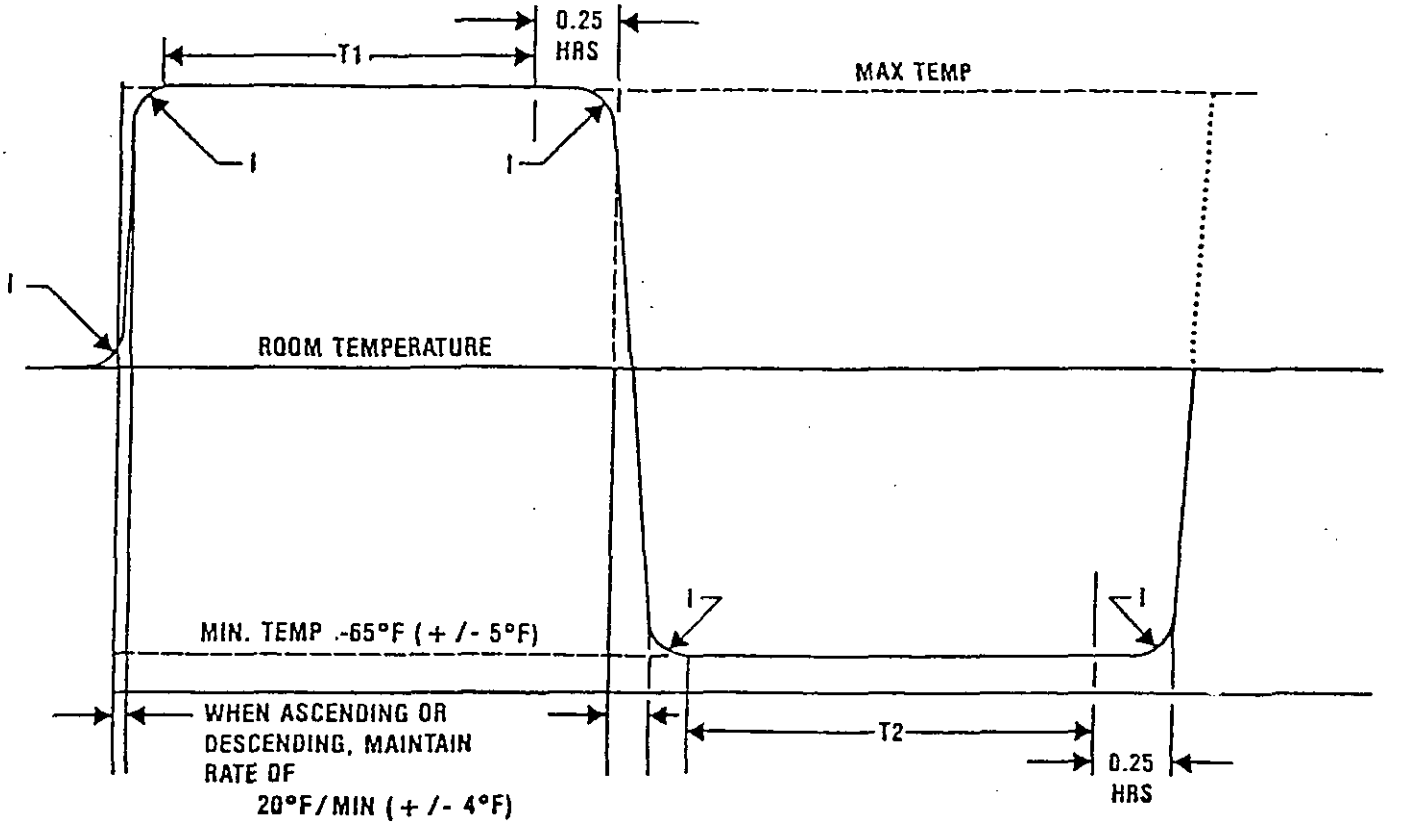
Generator voltage set point set at 28.1 volts direct current (vdc) (see 6.5.2); batteries connected; ambient temperature 65 to 85 degrees Fahrenheit (^{o}F); generator temperature stabilized unless specified. Unspecified tolerances are as follows:

a. Current: $\pm$ 10 amps

b. Voltage: $\pm$ 1 vdc

c. Speed, $\pm$ 500 revolutions per minute (rpm); temperature, $\pm$ 10°F.

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NOTE:

a. Regulator upper temperature = 250 + 0, -7.5°F maximum.

b. For a one-time determination of actual time required, connect a thermocouple to a regulator sample. The chamber shall be held constant at both high and low temperature extremes until the assembly temperature is within 5°F of the chamber temperature. Add 5 minutes to the time required to stabilize within 5°F; this time becomes T1 and T2.

c. Rounding off at upper and lower limits allowable (not to exceed 20 percent of ramp temperature).

FIGURE 1. Stress screen cycle.

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3.4.1 Generator current (see 4.6.1.1.). Generator current shall be as specified in table I. The generator current shall be enabled by 18 to 30 vdc on the excitation contact and disabled by removal of this voltage.

TABLE I. Generator current.

Test #	Rated Continuous current 1/ (amps)	Minimum voltage (vdc)	Maximum speed (rpm)	Generator air inlet temperature (°F)	Minimum test duration (hour)
1	100	25.2	1600	65 to 90	1
2	220	26.6	7000	65 to 90	1
3	150	26.6	7000	190 to 210	1

1/ Rated continuous output current must be produced at maximum speed. It may be produced at any lower speed.

NOTE: Maximum current is defined herein as current equivalent to or greater than the rated continuous output current under the conditions specified in table I.

3.4.2 Voltage set points (see 4.6.1.2). When operated at any speed greater than 1600 rpm with a load of 20 amps, generator output voltage shall be as follows for the three climatic conditions indicated on the voltage adjust control (see 6.5.2).

a. Arctic: 29.5 ± 0.3 vdc

b. Normal: 28.1 ± 0.3 vdc

c. Tropic: 27.4 ± 0.3 vdc

3.4.3 Voltage regulation (see 4.6.1.3). At speeds from 1600 to 8500 rpm and a load of 20 amps, voltage shall be within ± 0.6 vdc of the set voltage. At a speed of 7000 rpm and a load of 200 amps, voltage shall be within ± 0.8 vdc of the set voltage.

3.4.4 Voltage ripple (see 4.6.1.4).

a. With battery connected and measurements taken at battery terminals, peak-to-peak ripple voltage shall be not greater than 4.0 vdc at speeds from 1600 to 8500 rpm with a load of 20 amps.

b. With battery disconnected, voltage peaks including ripple shall be not greater than 40.0 vdc at speeds from 1600 to 8500 rpm with a load of 20 amps.

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3.4.5 Voltage transients (see 4.6.1.5). With or without battery on the line, voltage transients shall be within the limits specified in MIL-STD-1275. Switching to produce overshoot is defined as 187 ± 5 to 22 ± 2 amps in 20 ms maximum.

3.4.6 Efficiency (see 4.6.1.6). Minimum generator efficiency above 100 amps in the range 3000 to 7000 rpm shall be 60 percent. Torque required to drive the generator at any speed up to 8500 rpm shall be not greater than 30 foot pounds (ft lb).

3.4.7 Overspeed (see 4.6.1.7). Generator speeds up to 10,000 rpm for a period of up to 10 minutes shall not result in any electrical damage or impairment of generator performance.

3.5 Environmental conditions.

3.5.1 Low temperature (see 4.6.2.1). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation when stored and operated at temperatures down to -60 degrees Fahrenheit (^{o}F).

3.5.2 High temperature (see 4.6.2.2). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation when stored and operated at a temperature of 200°F.

3.5.3 Shock (see 4.6.2.3). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to sawtooth shock pulses of 40 gravity units (g) for 11 milliseconds (ms) time duration applied along each of the three major axes.

3.5.4 Vibration (see 4.6.2.4). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to vibration levels, in each of the three major axes, of 1.5 and 4.2 g in the frequency range of 5 to 500 to 5 Hz with a sweep time of 15 minutes.

3.5.5 Humidity (see 4.6.2.5). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation when exposed to a relative humidity ranging from 0 to 98 percent.

3.5.6 Steam and waterjet cleaning (see 4.6.2.6). The generator shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to steam cleaning, using high pressure steam and cleaner conforming to P-D-220, followed by waterjet cleaning.

3.5.7 Electromagnetic interference (EMI) (see 4.6.2.7). Electromagnetic interference shall be in accordance with MIL-STD-461, limits RE05 and CE07.

3.6 Vehicle system testing (see 4.7). Vehicle system tests shall be performed to assure mating system performance under vehicle operational environment and reliability.

3.6.1 Reliability (see 4.7.1). The generator shall demonstrate a minimum average life of 5,000 miles during vehicle testing.

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3.7 Nameplates and product marking.

3.7.1 Nameplates. Nameplate material and construction shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1. Nameplate shall be attached with suitable metal fasteners to achieve a permanent installation.

3.7.2 Product marking. Unless otherwise specified in the contract or order, the generator shall be identified in accordance with MIL-STD-130.

3.8 Workmanship. Workmanship shall be such as to provide a generator free from defects such as: cracked, broken, loose, or dented parts or components; scratched finish and/or peeled or blistered paint; burrs and/or sharp edges.

3.9 Nuclear survivability (see 4.8). This is a nuclear hardness critical item. Generator and all solid state (semiconductor) devices used in this assembly shall meet the nuclear survivability requirements specified in MRC-82304 MRC/SD-EP-82-027.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to determine conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Materials and manufacturing processes. The contractor shall maintain, and make available for Government review upon request, documentation verifying that materials and processes used in the manufacture of the generator conform to specified requirements. As a minimum, documentation shall confirm the following.

a. Materials, dissimilar metals and corrosion resistance in accordance with 3.2, 3.2.1 and 3.2.2.

b. Design and construction items as follows:

(1) Dimensions in accordance with 3.3.1.

(2) Protective coating in accordance with 3.3.3.

(3) Standard parts in accordance with 3.3.4.

(4) Power transistor in accordance with 3.3.5.

(5) Bearings in accordance with 3.3.6.

(6) Fan guard in accordance with 3.3.7.

(7) ESS in accordance with 3.3.8.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: $77^{o}F \pm 18^{o}F$

b. Relative humidity: uncontrolled room ambient

c. Atmospheric pressure: site pressure

4.4 First article inspection.

4.4.1 Preproduction inspection (see 3.1.1). Two generators produced in accordance with 3.1.1 shall be subjected to the preproduction inspection (see 6.3.1).

4.4.1.1 Preproduction inspection routine. The preproduction samples shall be subjected to the inspections specified in table II.

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4.4.1.2 Preproduction failure. Inability of any generator to pass any of the tests specified in table II shall be deemed a failure. In the event of a failure, the failed part or parts shall be returned to the contractor. The contractor shall perform a failure analysis of each part and submit a corrective action plan of sufficient validity to justify design acceptability without retest. The Government shall approve the corrective action plan and any retest activity.

4.4.1.3 Design change. Whenever a change is made in the design, the Government shall be notified to determine whether the supplier must perform any or all of the preproduction inspections specified herein.

4.4.1.4 Disposition of preproduction samples. Samples which have been subjected to preproduction inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new equipment under the contract.

4.4.2 Initial production inspection (see 3.1.2). Three generators produced in accordance with 3.1.2 shall be subjected to initial production inspection.

4.4.2.1 Initial production inspection routine. The initial production samples shall be subjected to the inspections specified in table II.

4.4.2.2 Initial production failure. Inability of any of the initial production samples to pass any examination or test shall constitute a failure. Failure of a sample to pass any examination or test shall be cause for refusal to grant initial production approval. Such failure may, at the option of the Government, be cause for refusal to allow additional examinations or tests until causes of faults have been corrected and the corrective action approved by the Government.

4.4.2.3 Retention of initial production approval. Unless extended by the Government to other contracts, initial production approval is valid only on the contract under which it is granted.

4.4.2.3.1 Changes to processes, procedures, or location. Whenever a change is made in the design or the manufacturing process, procedures, or facility location used in the production of generators, the Government shall be notified to determine whether the contractor must requalify for initial production approval.

4.4.2.4 Disposition of initial production samples. Samples which have been subjected to initial production inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new equipment under the contract.

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TABLE II. First article inspection.

Inspection	Requirement Paragraph	Inspection Method	Pre-production Sample		Initial Production Sample		
			1	2	1	2	3
100 percent	table III	table III	X	X	X	X	X
Weight	3.3.2	Scale	X	X	X	X	X
Gen. current (tests 1 & 3 of table I)	3.4.1	4.6.1.1	X	X	X	X	X
Voltage set points	3.4.2	4.6.1.2	X	X	X	X	X
Voltage regulation	3.4.3	4.6.1.3	X	X	X	X	X
Voltage ripple	3.4.4	4.6.1.4	X	X	X	X	X
Voltage transients	3.4.5	4.6.1.5	X	X	X	X	X
Efficiency	3.4.6	4.6.1.6	X	X	X	X	X
Overspeed	3.4.7	4.6.1.7	X	X	X	X	X
Low temperature	3.5.1	4.6.2.1	X	X	X	X	X
High temperature	3.5.2	4.6.2.2	X	X	X	X	X
Shock	3.5.3	4.6.2.3	X	X	X	X	X
Vibration	3.5.4	4.6.2.4	X	X	X	X	X
Humidity	3.5.5	4.6.2.5	X	X	X	X	X
Steam and waterjet cleaning	3.5.6	4.6.2.6	X	X	X	X	X
EMI	3.5.7	4.6.2.7	X	X			
Vehicle system testing	3.6	4.7	X	X			
Nuclear survivability	3.9	4.8	X	X			

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4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of 100 percent inspection.

4.5.1.1 100 percent inspection. Every generator shall be subjected to the inspections specified in table III.

4.5.1.1.1 Failure. The supplier may rework a rejected generator to correct the defect(s) and resubmit it for inspection. Rework shall be within the confines of drawing and specification requirements.

TABLE III. 100 percent inspection.

Inspection	Requirement Paragraph	Inspection Method
Generator current	3.4.1 (test 2 only)	4.6.1.1.2
Identification marking	3.7	Visual
Workmanship	3.8	Visual

4.5.2 Inspection of packaging. The sampling and inspection of packaging shall be in accordance with 5.1.

4.6 Methods of inspection. The following methods of inspection are acceptable for verifying compliance with the requirements of 3.4 and 3.5.

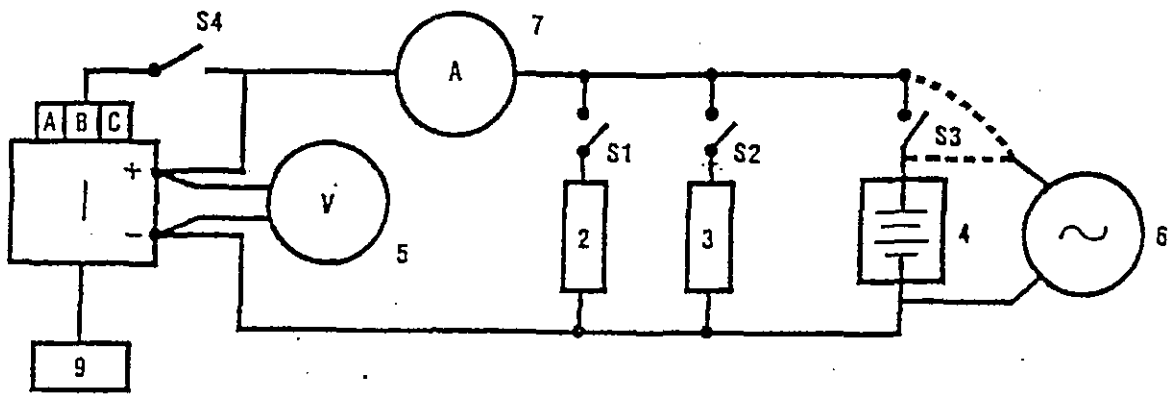
4.6.1 Performance tests.

4.6.1.1 Generator current (see 3.4.1).

4.6.1.1.1 Generator current for first article testing. The generator shall be connected to the test circuit as shown in figure 2 with the voltage adjustment set for 28.1 vdc. Close switches S1, S3 and S4 to connect battery and load. Generator shall be operated for a period of one hour minimum at each of the specified speeds and temperature conditions of table I. Measure and record output current and verify that it meets values specified in table I.

4.6.1.1.2 Generator current for 100 percent inspection. The generator shall be connected to the test circuit as shown in figure 2 with the voltage adjustment set for 28.1 vdc. Close switches S1, S3 and S4 to connect battery and load. Perform test 2, table I; measure and record output current and verify that it meets values specified in table I.

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LEGEND:

1. Generator under test
2. Adjustable resistive load bank
3. Auxiliary load bank
4. Battery 200 to 240AH fully charged
5. Voltmeter (4 digit minimum)
6. Oscilloscope
7. Ammeter (400 A)
8. S1, S2, S3, S4 switches
9. Tach (motor/rpm monitor)

FIGURE 2. Generator Test Set-Up

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4.6.1.2 Voltage set points (see 3.4.2). The generator shall be connected to the test circuit as shown in figure 2. Close switches S1, S3 and S4 to connect battery and load. Verify that performance requirements specified in 3.4.2 have been met.

4.6.1.3 Voltage regulation (see 3.4.3). The generator shall be connected to the test circuit shown in figure 2. Close switches S1, S3, and S4 to energize the generator and connect battery and load. Perform testing as follows:

a. Operate the generator at 2000 ± 400 rpm at 20 amps. Verify that output voltage is within ± 0.6 vdc of the set voltage. Disconnect the battery (switch S3).

b. Repeat step a. at 4000 ± 400 rpm, 6000 ± 400 rpm, and 7250 ± 250 rpm.

c. Repeat step a. at 7250 ± 250 rpm at 200 amps. Verify that output voltage is within 0.8 vdc of the set point.

4.6.1.4 Voltage ripple (see 3.4.4).

a. The generator shall be connected to the test circuit shown in figure 2. Close switches S1, S3 and S4 to energize the generator and connect battery and load. With the generator operating as specified in 4.6.1.3 a. and b., peak-to-peak ripple voltage shall not exceed 4.0 vdc when measured with an oscilloscope at the battery terminals.

b. Disconnect the battery (open switch S3) and connect the oscilloscope to the output terminals of the generator. Peak output voltage including ripple shall not exceed 40 vdc.

4.6.1.5 Voltage transients (see 3.4.5). The generator shall be connected to the test circuit shown in figure 2. Close switches S2, S3 and S4 to energize the generator and to connect an auxiliary load of 22 ± 2 amps. Close switch S1 and adjust the main load for a total current of 187 ± 5 amps. Open switch S3. Open switch S1 and measure the voltage transient on the oscilloscope. The voltage transients shall be within the loci specified in MIL-STD-1275. Switching time (arc duration) as observed on the oscilloscope shall not exceed 20 ms. A spring loaded knife switch shall be used. Capacitors to adjust the switching time are not allowed. Repeat the procedure, except close switch S3 to connect the battery. A fully charged battery shall be used (battery shall draw less than 5 amps from a 28 vdc charging source). All surges and spikes shall be within the loci specified in MIL-STD-1275. Repeat the tests by switching the load from 22 ± 2 amps to 187 ± 5 amps. All surges and spikes shall be within the loci of MIL-STD-1275. The voltage transient tests shall be performed at a generator speed of 7000 to 7500 rpm.

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4.6.1.6 Efficiency (see 3.4.6). The generator input torque and rpm shall be measured between 65 and 85°F ambient temperature while the generator produces maximum current at 28.0 vdc. For this test, the voltage adjustment shall be in the arctic (29.5 ± 0.3 vdc) position. Torque and current shall be measured in the speed range from 2000 to 8500 rpm and recorded after the internal machine temperature has stabilized (less than 1 amp output current change in 5 minutes).

Generator input power (P [W_i]): $$P\,[W_i] = \frac{\text{Torque (inch lbs.)}}{84.6} * \text{rpm}$$

Generator output power (P [W_o]): $$P\,[W_o] = I(\text{out}) * V(\text{out})$$

$$\text{Efficiency [\%]} = \frac{P\ (\text{output})}{P\ (\text{input})} * 100$$

Verify that peak torque does not exceed 30 ft lbs and the efficiency meets the requirements of 3.4.6.

4.6.1.7 Overspeed (see 3.4.7). The generator shall be operated for a minimum of 60 seconds at a speed of 10,000 ± 500 rpm with a load of 20 amps. There shall be no indication of any damage while monitoring the output voltage. Then adjust the load to 187 ± 5 and operate the generator at 7000 to 7500 rpm. Compare the output voltage with the value obtained in 4.6.1.3 a. and b. to verify that there is no impairment of generator performance.

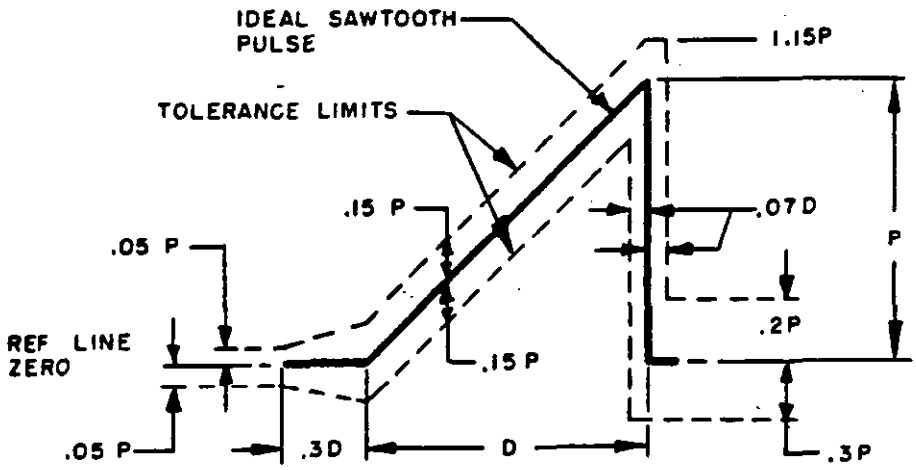
4.6.2 Environmental tests. The following methods of inspection are acceptable for verifying compliance with 3.5. Prior to and at the conclusion of each of the environmental tests, the generator shall be operated under stabilized standard ambient conditions and examined for evidence of deformation or damage.

4.6.2.1 Low temperature (see 3.5.1). The generator shall be placed in a temperature chamber in a manner simulating service usage. Lower the chamber temperature to -60°F and maintain for a period of 24 hours after test temperature stabilization (see 6.5.4). Conformance to the performance requirements specified in 3.4.1, test #2 shall be verified before and within 5 minutes after low temperature exposure.

4.6.2.2 High temperature (see 3.5.2). The generator shall be placed in a temperature chamber in a manner simulating service usage. Raise the chamber temperature to 200°F and maintain for a period of 48 hours with a relative humidity not greater than 15 percent. Conformance to the performance requirements specified in 3.4.1, test #2 shall be verified before and after high temperature exposure, and while the chamber is at the highest operating temperature.

4.6.2.3 Shock (see 3.5.3). The generator shall be mounted in a manner that simulates actual use, or shall be rigidly mounted to a shock table, and subjected to three sawtooth shocks of 40g and time duration of 11 ms (see figure 3) along the three mutually perpendicular axes of the generator for a total of 18 shocks. No performance testing shall be required during shock. Conformance to the performance requirements specified in 3.4.1, test #2 shall be verified prior to and at the conclusion of shock.

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P = Peak value = 40 g
D = Nominal duration = 11ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries, and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$, where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 3. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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4.6.2.4 Vibration (see 3.5.4). The generator shall be mounted in a test fixture and subjected to the vibration test as specified in 3.5.4 and figure 4. During vibration testing the generator shall be operated at a single speed between 2000 and 7000 rpm with a load of 20 amps. Also during vibration, the voltage shall be within ± 0.6 vdc of the set voltage.

4.6.2.5 Humidity (see 3.5.5).

a. The generator shall be placed in a chamber trap-vented to the atmosphere to prevent the build up of total pressure. No condensation shall be permitted to drip on the generator. Relative humidity shall be determined from dry bulb/wet bulb thermometer comparison method or an equivalent method approved by the Government. Air velocity flowing across the wet bulb shall be not less than 900 feet per minute (ft/min). Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall be not greater than 150 ft/min. Steam or distilled, demineralized, or deionized water having a hydrogen ion concentration (PH) between 6.0 and 7.2 at 73°F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the generator by the test facility.

b. After having been dried for 24 hours at 129 ± 3°F, the generator shall be conditioned at 73 ± 3°F and 50 ± 10 percent relative humidity for 24 hours.

c. Raise the internal chamber temperature to 86 ± 3°F and relative humidity to 94 ± 4 percent. Expose the generator to 5 continuous 48-hour cycles in accordance with figure 5. Accumulated moisture may be removed by turning the generator upside down or shaking. Wiping is not permitted. No operational tests are required during humidity testing. After completion of cycling, condition the generator for 24 hours at 73 ± 3°F and 50 ± 10 percent relative humidity. Immediately upon removal of generator from humidity chamber, it shall be placed in a moisture-proof container until beginning of performance tests. Within one hour after humidity testing, the generator shall be subjected to test #2 of 3.4.1.

4.6.2.6 Steam and waterjet cleaning (see 3.5.6). The generator shall be subjected to steam cleaning, immediately followed by waterjet cleaning as follows. The jet shall be applied perpendicular to the surface being cleaned at a distance of not greater than 1 foot for steam cleaning, and not greater than 3 feet for waterjet cleaning. The jet pressure shall be 105 ± 5 pound-force per square inch gage (psig) for steam cleaning and 50 ± 5 psig for waterjet cleaning. The jet shall be moved at such a rate that the exposed surface of the generator is subjected to the jet at the rate of 1 square foot per minute for 2 to 3 minutes. Prior to and within 1 hour (unless otherwise directed by the Government) after completing the steam and waterjet cleaning, the generator shall be subjected to test #2 of 3.4.1. There shall be no entry of water or moisture into modules sealed against water or moisture. Moisture inside the generator shall not be considered a failure.

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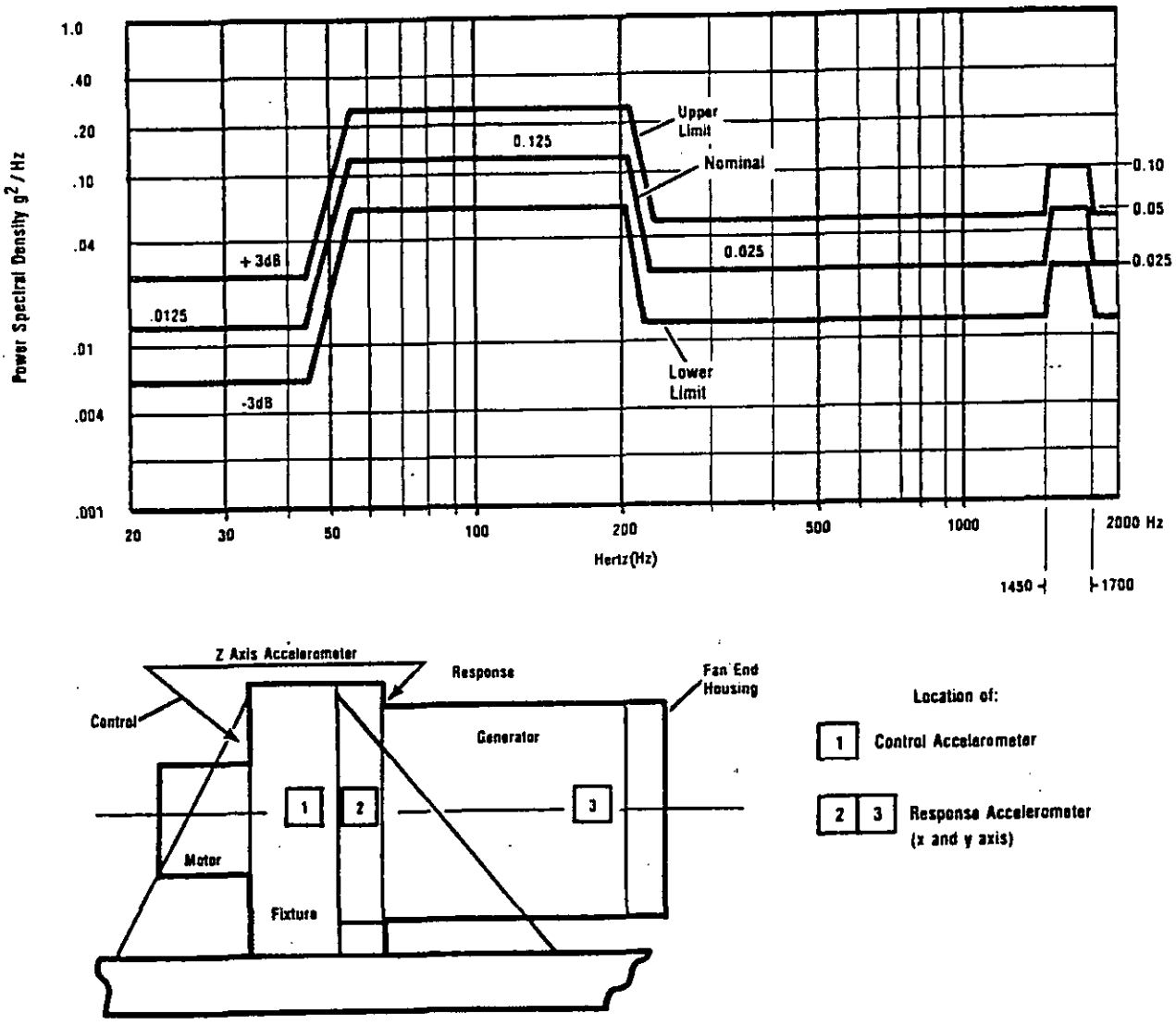
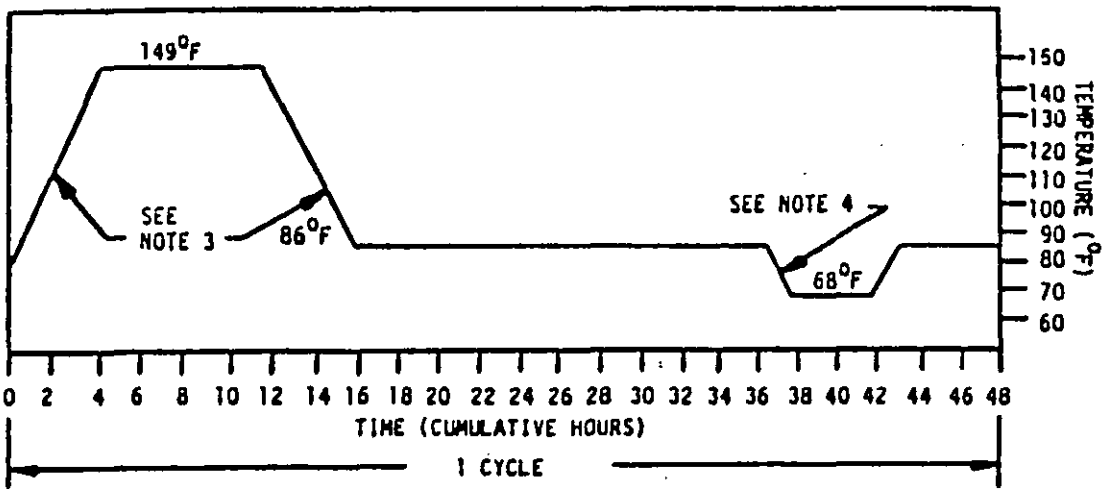


FIGURE 4. Generator random vibration.

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NOTES:
1. Tolerance during temperature change shall be not greater than 5°F.
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86 and 149°F shall be not less than 14.4°F per hour.
4. The temperature increase in this portion of the curve shall be not less than 18°F.

FIGURE 5. Humidity cycle.

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4.6.2.7 EMI (see 3.5.7). The generator shall be tested in accordance with MIL-STD-462. Conducted and radiated measurement shall be made at 2300 to 2400 rpm and 7000 to 7500 rpm and specified loads. Perform the test at standard ambient temperature. The machine temperature need not be stabilized. The voltage adjust shall be in the normal position. At 2300 to 2400 rpm, the generator output current shall be 60 amps, 120 amps, and a maximum current at a generator terminal voltage of 27.0 ± 0.3 vdc. At 7000 to 7500 rpm, the load shall be 150 amps, 200 amps, and a maximum current at a generator terminal voltage of 27.0 ± 0.3 vdc. Verify that the conducted and radiated emissions do not exceed the CE07 and RE05 levels in accordance with MIL-STD-461.

4.7 Vehicle system testing (see 3.6). To determine conformance to 3.6, generators submitted for vehicle system testing shall have been subjected to all preproduction inspections identified in table II prior to the start of vehicle tests. Three generators will be used for vehicle system testing (one of which will be used as a spare). Two generators shall be installed on test vehicles which will be operated for 6000 miles each under the conditions specified in the product function specification for the vehicle. After completion of vehicle testing, each generator shall be subjected to the generator current test specified in table III. Vehicle testing shall be the responsibility of the procuring activity; whereas, all other inspections specified herein shall be the responsibility of the supplier. Reliability shall be as specified in 4.7.1.

4.7.1 Reliability (see 3.6.1). Minimum acceptable reliability as demonstrated during vehicle testing is 5000 mean miles between failure (MMBF). Reliability calculation will be a point estimate based on total vehicle test miles divided by the total chargeable failures attributed to the generator. Failure is defined as generator replacement required because of inability to perform its required function.

4.8 Nuclear survivability (see 3.9). Conformance to the requirements of 3.9 shall be verified through analyses and tests to be performed by the procuring activity.

5. PACKAGING

5.1 Preservation, packaging and marking. Packaging for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Generators manufactured in accordance with this specification are intended for use on military tracked vehicles in the Bradley Fighting Vehicle (BFV) Family.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, revision, and date of this specification.

b. Title, number, revision, and date of part drawing.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

d. Preproduction inspection, if required (see 3.1.1).

e. Initial production inspection, if required (see 3.1.2).

f. ESS, if other than as specified herein (see 3.3.8).

g. Responsibility for inspection, if other than as specified herein (see 4.1).

h. Level of packaging, if other than as specified herein (see 5.1).

i. Any deviation from this specification.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to bidders whether it shall include preproduction samples and/or initial production samples (see 3.1). The acquisition document should include specific instructions regarding the arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive first article inspection, or any portion thereof, to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products should furnish evidence with the bid that prior first article approval is presently appropriate for the pending contract.

6.3.1 Preproduction inspection. Preproduction inspection should be performed prior to initiation of full scale production or rate production of generators. Preproduction is only required for units being produced for the first time by new suppliers. A supplier complying with the requirements of preproduction inspection should be considered a qualified source and should be identified on Drawing 19207-12295268. Bradley Fighting Vehicle System Program Manager Office (BFVS PMO) should be responsible for maintaining records of qualified sources.

6.4 Definitions.

6.4.1 Environmental stress screening. Environmental stress screening (ESS) is a process that subjects devices to physical and climatic stress (or combinations thereof). This stress forces flaws that are not ordinarily apparent into observable failures which can then be removed. Flaws are defects in design, workmanship or material that will eventually result in the failure of the device prior to its normal life expectancy.

6.4.2 Test temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary more than 3.6°F per hour.

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6.4.3 Voltage adjust control. The generator is provided with a three-step adjustment feature. This feature provides for changing the output voltage to compensate for battery charging current under arctic, normal or tropical conditions. The adjustment is applied as shown in figure 6. The nominal voltage set points are as specified in 3.4.2.

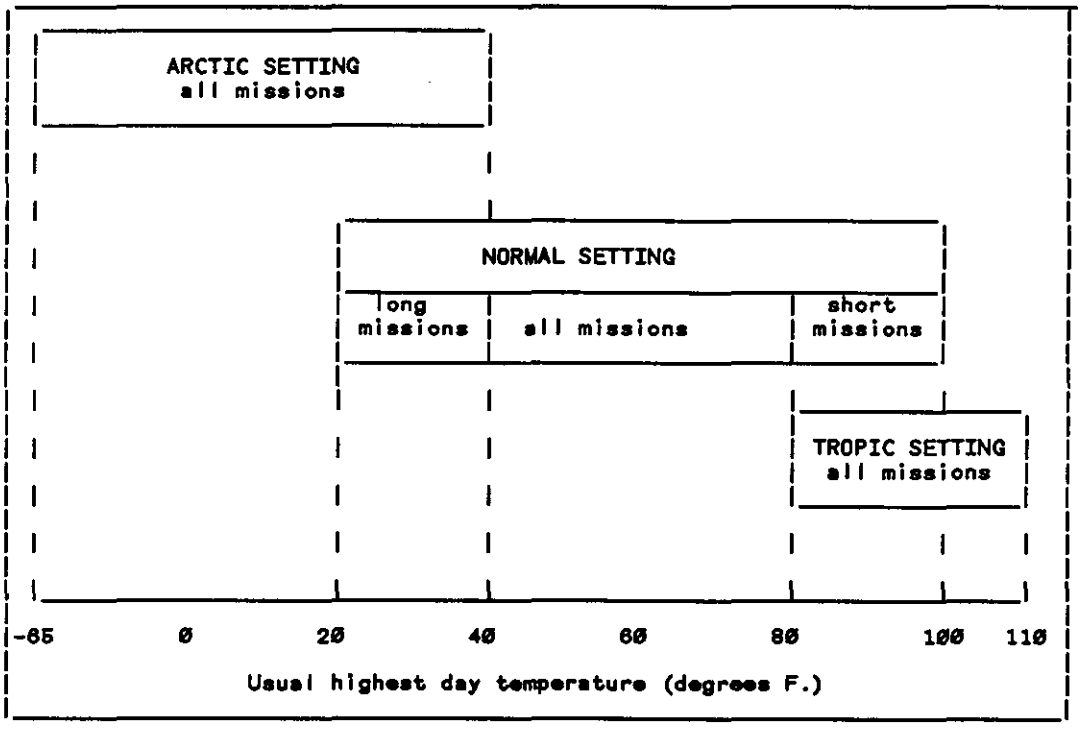


FIGURE 6. Voltage adjust.

6.5 Supersession data. This specification supersedes MIL-G-62496A(AT), dated 22 June 1989 and FMC Drawing 19207-12292165 revision F, dated 03 June 1988.

6.6 ESS of the regulator. The following procedure is recommended for environmental stress screening of the regulator.

a. Install the regulator into the temperature chamber, and perform the following steps:

(1) Apply 30 +1, -0 vdc to the regulator terminal (B+) and ground. Install a 100-ohm ± 10 percent, 10-watt, resistor from B+ to the collector of the field power transistor.

(2) Monitor the current drain at the first and last stress screening cycles at maximum temperature.

(3) Perform 15 stress screen cycles as shown in figure 1. Maximum temperature shall be 250 +0, -7.5°F, minimum temperature shall be -65 ± 5°F.

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(4) Disconnect power and remove regulators from the chamber.

(5) Compare the current drain of the first cycle to the last cycle. Verify that the current drain of the last cycle is within 25 percent of the current drain of the first. If the assembly passes the test, mark the regulator with an indelible yellow dot.

b. If failure occurs on any component or interconnection, failed components shall be replaced with stress screened components. Defective connections shall be repaired as required. Rescreening of the assembly is not required. The assembly shall be retested per step a(5). If the assembly passes the test, mark the regulator with an indelible yellow dot.

6.7 Subject term (key word) listing.

Power transistor
Power input
Voltage: regulation, adjustment, ripple, transients

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2920-A017)

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	MIL-G-62496B(AT)	92/07/31

3. DOCUMENT TITLE

Generator, Engine Accessory, 220 AMP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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U.S. Army Tank-Automotive Command	(313) 574-5413	786-5413
c. ADDRESS (Include Zip Code) ATTN: AMSTA-GDSM Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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